EVALUATION OF UNDERGRADUATE USE OF SELECTED SOCIAL NETWORKS FOR INSTRUCTION IN LAGOS STATE

INTRODUCTION

1.1 BACKGROUND OF THE STUDY

The advent of information and communication technology now allows for easy access to information in a convenient form. Social networks such as Zoom, Google meet, Microsoft teams, Edmondo, enable easier undergraduate instruction from anywhere and at any time in the world. Through these apps, information is becoming intertwined with our daily lives and could either enhance productivity, efficiency and intelligence or make users vulnerable to its side effects (Fauzi, 2019). Using social media such as Facebook, WhatsApp, Instagram, Zoom, Skype, Webinars, and so on provides free access to online communication and information to undergraduates. Through these various platforms, teachers can connect with students and incorporate social media into their lessons, making them more interesting, relatable and engaging (Willbold, 2019). Today, both social networks and technology have become an indispensable part of our lives.

Social networks have gradually emerged as a new avenue of communication between teachers and students, mainly in higher education, becoming an important communicative tools (Akcaoglu & Bowman, 2016; Albayrak & Yildirim, 2015), generating diverse questions about teacher–student communication through social networks and their impact on the teaching–learning process (Hershkovitz & Forkosh-Baruch, 2013). The use of social networks provides the opportunity for teachers and students to be in continuous contact transcending the conventional classroom and creating new teaching and learning spaces (Ean & Lee, 2016). However, this new reality has generated debates

as to whether teachers and students should interact in this way, leading some authorities to restrict or even prohibit such communication (Asterhan & Rosenberg, 2015). Supporters of the teacher–student interaction through social networks present arguments in favor, such as teachers' and students' freedom of expression, the inevitability of the phenomenon and the pedagogical potential of social networks (Greenhow et al., 2009).

Social networks are great tools when it comes to interacting with students since it's something they are so familiar with, using these sites frequently throughout daily life (Segaren, 2019). In a university setting, social media can be useful when it comes to reaching out and engaging with students (Segaren, 2019). It's a less formal way for lecturers to communicate, showing students a more personal side while improving the teacher–student relationship. It will not be wrong to say that social media and education goes hand in hand in the modern world. However, the question that arises is: how is social media useful to students? Are there ways that social media in education can benefit learners and help to fill in the gaps in teaching and learning? Jones (2015) wonders on how lecturers might shift from 'uploading' content to students to a better teaching technique that allows students to participate more enthusiastically in the teaching and learning process i.e. learner-centred approach.

The current COVID-19 and advances in information and communication technologies (ICTs) necessitated the reconsideration of alternative means of teaching and learning by students and educators. As a result, the use of social networks such as Zoom, Google meet, Microsoft Teams has been adopted by professionals to teach, learn and relearn in this challenging times as all students are at home.

Various countries have adopted different strategies for teaching and learning in the bid to assist students who are out-of-school within the period the schools would remain closed. This calls for remote learning in order not to put teaching and learning in

a halt. The integration of educational technology (EdTech) to deliver and support remote learning in almost all cases are being adopted by most schools. The systems of learning online such as (Learning Management Systems, Video Conferencing) are mostly deployed by middle-and high-income environments while some are adopting other broadcast media like television for delivery as a means of supplementary. Within countries, it is however revealed that the demand for distant learning has also made clearly unambiguous digital divides (World Bank, 2020).

The effects of modern information technology are increasing rapidly, and the ICTs are part of daily education activity especially among undergraduates universities, especially in the education process. New innovative technologies such as Microsoft Teams and Zoom, among others, become useful and effective (Seyal, Mohd, Awg, Yussof & Rahman, 2017).This means that students can attend classes, anytime it is fixed, anywhere because they do not have to attend a physical class. Students can also learn face-to-face via a variety of channels (TV, internet), tutors, classrooms (Bielaczyc, 2006 as cited in Nassoura, 2012). Over time, ICT systems, Microsoft Teams, Moodle Platforms and other Learning Management Systems (LMS), have been adopted by academics, institutes of higher education, and secondary schools, especially during this Covid19 pandemic period. This has resulted in diversified education system incorporating remote learning and online courses and tests, as well as a combination of the usual one-on-one lectures and tutoring. Universities and secondary schools in Nigeria have invested in the provision of resources and technology to prevent disruption to teaching and learning while restrictions due to the pandemic last. During this COVID-19 pandemic, most concerns in the educational sector about remote learning have focused on how the instructors/teachers incorporate new technology in their teaching in order to engage students at their various homes. Microsoft teams, Moodle, and Zoom are some of the ICT platforms through `which students are taught during the pandemic. Microsoft Teams is a collaborative platform complete with document sharing, online

meetings, and many more features which could be useful for remote learning. Remote learning occurs when the learner and instructor are separated by distance, therefore cannot meet in a traditional classroom setting. It is an effective way to reach students with resources that would otherwise be unreachable during the school shut down, through a flexible, customizable accessible learning space. A few of the goals of remote learning are to streamline the process for continued learning and to keep school communities connected. Although remote learning has the same social dynamics as online delivery its implications are different in that learners rely on the facilitator to bridge the physical distance that separates the participants and ensure that the contents don't feel static and distance to students. Therefore, some of the considerations when working with students remotely would be to use web applications that provide synchronous and asynchronous engagements. Some other factors to consider would include bandwidth access, computer access, and students' emotional conditions.

Microsoft Teams refers to a digital cloud app hub that puts together discussions, meetings, files, and applications in a single LMS (Learning Management System) (Microsoft, 2018). Tsai (2018) expects Microsoft Teams to experience the fastest growth in the next two years. 41 percent of organisations plan to use Microsoft Teams globally by the end of 2020, while Microsoft Team has 75 million users by April 2020. Microsoft Teams has been adopted by various schools to streamline the remote learning process. The use of Microsoft Teams has been said to greatly enhance teaching and learning, teachers' ability to grade and monitor students' assignment, classroom organization, and teacher-student interaction (Alameri, Masadeh, Hamadallah, Ismail & Fakhouri, 2020). Effective learning with the use of Microsoft Teams requires that teachers and students be trained in it. Lu (2010) also noted that teachers and student's behaviour using Microsoft Teams platforms could be influenced by self-efficiency, age, training, and internet availability.

Also, physical and psychological factors in the Microsoft Teams environment could either encourage or inhibit the students' attitude to and performance in learning (Zandvliet, 2003 as cited in Adewole-Odeshi, 2014). These factors and other social factors in remote learning impact on the values which students and teachers share in education, and they should therefore also be considered alongside the technological tools.

The functional domains required in using Microsoft Teams could be grouped into affective, cognitive, and psychomotor. The affective domain encompasses the students and teachers' feelings, emotions, and attitudes concerning Microsoft Teams. This depicts how they feel about certain Microsoft Teams capabilities. Assignments, screen sharing, white boarding, immersive reading, together mode, LMS integration, insight, and so on are just a few of these tools and capabilities available on Microsoft Teams. The cognitive aspect refers to the academic discourse engagement between the students and teachers when using the Microsoft Teams platform for acquisition of knowledge and, the final domain, psychomotor, covers teachers and students' motor skills as well as their ability to coordinate them for efficient Microsoft Teams engagement. The nature of appraisal and engagement with these functional aspects of the Microsoft Teams platform could significantly inhibit or enhance remote learning (Alabdullaziz, Muhammad, Alyahya & James, 2011). Even though studies have been done on teacher's perceptions of remote learning, there are not many on the use of Microsoft Teams for remote learning by teachers in Nigeria.

Several studies have explored the use of different social platforms in Nigeria, and yet, only a limited number of studies have focused on these platforms for teaching and learning purposes. Therefore, this study aims to investigate undergraduate use of social network like Zoom, Google meet, Microsoft Teams for instruction in Lagos State.

1.2 STATEMENT OF THE PROBLEM

Since the current COVID-19 and advances in information and communication technologies (ICTs) the reconsideration of alternative means of instruction is necessary for undergraduates.

Several studies have explored the use of different social networks in Nigeria, and yet, only a limited number of undergraduates have focused on these platforms for instruction.

The use of social networks such as Zoom, Google meet, Microsoft Teams has been adopted by professionals in business and industry in this challenging times by but undergraduates in Nigeria have Lagos adopted to its use for instruction.

The social media has been a very tool for communication and instruction

This study aims to investigate undergraduate use of social network like Zoom, Google meet, Microsoft Teams for instruction in Lagos State.

Therefore it becomes necessary to ask, to what extent are undergraduates aware, adopt, frequently use, understand the relevance of social networks like Zoom, Microsoft Teams, Google Meet for instruction.

1.3 PURPOSE OF THE STUDY

The purpose of the study is investigate undergraduate use of social network like Zoom, Google meet, Microsoft Teams for instruction in Lagos State.

Specifically, the study intends to find out:

1. Undergraduate awareness about the availability of social networks for instruction
2. Various social networks undergraduate use for instruction

3. The frequency of use of available social network for undergraduate instruction
4. Undergraduates understanding of the functions in using social networks for instruction
5. Undergraduate challenges in using of social networks for instruction

1.4 RESEARCH QUESTIONS

The following research questions have been formulated to guide the study.

1. Are undergraduate aware about the availability of social networks for instruction?
2. What social networks do undergraduate uses for instruction?
3. How frequent do undergraduates use social network for instruction?
4. Do undergraduates understand the functions and use of social networks in instruction?
5. What are the challenges undergraduate face in using of social networks for instruction?

1.5 SIGNIFICANCE OF THE STUDY

Practically the following will benefit from the findings of this study, undergraduates, curriculum developers, the government, lecturers, higher institutions, Educational guidance and counselors etc.

1.6 SCOPE AND DELIMITATION OF THE STUDY

The Scope has limited the investigation to undergraduate of tertiary institutions in Lagos state.

The study will be limited to undergraduate from selected tertiary institutions in Lagos state.

CHAPTER TWO

LITERATURE REVIEW

Literature is replete with previous studies undertaken to demystify the use of social media in classroom environments of higher education. In the subsequent sections, the authors reviewed and synthesize some of the most potent studies that relate quite well with the current study.

2.1 USE OF SOCIAL MEDIA PLATFORMS IN EDUCATION

The advancement in technological innovation is increasingly influencing the way information is communicated through diverse channels particularly the social media and these changes are also shaping the teaching–learning process in the educational system. Social networking sites as applications for smartphones, android phones enable individuals to connect from any location to Facebook, Instagram, YouTube, LinkedIn or WhatsApp without having to connect to the site (Magro et al., 2013). Lim See Yin et al. (2014) investigated undergraduate informatics students' engagement with social media for academic purpose in Malaysia. The results revealed a close match of ownership, time spent online, types of social media users, and pattern of usage between informatics and non-informatics students. The study also indicated that students and lecturers had accepted and explored the potency of social media for engagement with the institution, their peers, as well as for teaching and learning purposes. Similarly, Froment et al. (2017) analyzed the use of social networks as a communication tool between teachers and found that they primarily communicate through social networks for academic reasons. The study revealed that teachers found Facebook and WhatsApp to be the most effective social media tools due to the popularity and versatility of Facebook, and the easy-to-use communicative abilities of WhatsApp. For students taking distance learning courses or for lectures with large class sizes, teachers can make

themselves more accessible to students via social media. They can hold 'virtual office hours' through Skype or get them to use Facebook, Twitter or WhatsApp to raise questions they didn't get the chance to in class (Froment et al., 2017). Froment et al. (2017) concluded that teachers should use social networks as communication tools for purely academic-related issues, such as addressing aspects related to class organization and coursework. The study by Chawinga (2017) incorporated Twitter and blogs into two undergraduate courses offered in the Department of Library and Information Science at Mzuzu University, a public university in Malawi. Results suggest that if appropriately deployed, Twitter and blogs are catalysts for the much hyped learner-centred approach to teaching because using these technologies, it emerged that students shared and discussed course materials, posted their course reflections and interacted amongst themselves and with their lecturer 24/7.

WhatsApp is a primary mobile instant messaging application that enables users to send each other text, images, video and audio messages. It allows individuals to socialize and stay connected with Internet connection available. Several studies found that using WhatsApp was an efficient communication technology and concluded that it was a useful tool that can transfer data rapidly (Dorwal et al., 2016; Gulacti et al., 2016). Though there are lots of instant messaging applications that can operate on mobile devices, it seems that WhatsApp application is one of the most favored mobile based applications (see SimilarWeb, 2016; Statista, 2016). In their study, Church and de Oliveira (2013) emphasized this fact and stated that WhatsApp has grown in popularity due to its benefits such as, being able to send real-time messages to an individual or groups of friends simultaneously, low-cost, and privacy. WhatsApp development as a discussion group has been carried out by several researchers. Results from several studies have showed support for the use of WhatsApp as a discussion group (Martinez, 2017; Willemse, 2015).

WhatsApp Group Discussion is a WhatsApp application feature to form a group that discusses a particular topic. One of the special characteristics of WhatsApp is the capability to create and interact with groups from 3 to 256 members. Studies have shown that almost all students already have the WhatsApp application installed on their smartphone, and WhatsApp is the most commonly used application because it makes it easy for users and shortens the time to get information (Baro et al., 2015; Semode et al., 2017; Willemse, 2015). Fauzi (2019) conducted a study on the development of WhatsApp group discussion to solve the limitation of lecturer–students' interaction at class in Universitas Nahdlatul Ulama Blitar (UNU), Indonesia. The research based on the idea of overcoming problems about the limited interaction between teachers and students in the classroom, and responding to technological developments that enter education. The study produced a WhatsApp Group Discussion product that can be used in lectures. The results of the study showed that in combination, the product developed have met the criteria sufficiently valid. This indicates that in terms of learning and technology, the product (WhatsApp Group Discussion) is feasible and can be used in lectures. This product can be used to overcome the problem of limited interaction between the lecturers and the students of UNU Blitar in the lecture class (Fauzi, 2019).

Agbo et al. (2020) investigated students' perceptions and opinions regarding the use of WhatsApp social media group for learning computing education in the Nigerian context. The study explores how social media usage by students influences their perceived learning outcomes. The study found a positive relationship between social media usage for computing education and perceived learning outcomes. In addition, the study found a linear relationship between communication in group and perceived learning outcomes. Similarly, Cetinkaya (2017) explored the effects of WhatsApp use for education and determine the opinions of students toward the process. The analysis indicated that both learning environments have different effects on the success of students and that supporting the traditional environment by using WhatsApp is more effective. The

qualitative analysis showed that students developed positive opinions toward the use of WhatsApp in their courses. Respondents reported that learning could also take place unconsciously and the messages with images were more effective for their learning.

2.2 BENEFITS OF USING SOCIAL NETWORKS IN INSTRUCTION

In recent times, social media such as Telegram, WhatsApp, Zoom, and Facebook have been used for collaborative learning and create engagement among a network of students, thus making these integral media part of students' social and academic life (Agbo et al., 2020). The social media by their nature have the capabilities of educating, informing, entertaining and inflaming the audience, of which are also the basic function of the mass media. Above all, they possess a contagious and outreaching influence which the conventional media lack (Agbo et al., 2020). This potential is most likely what Osahenye (2011) referred to as "unstoppable power of the social media". The social media and social network tools especially WhatsApp Messenger, Facebook, and Viber, Skype, have become dominant factor in today's digital world and are affecting how users communicate and businesses operate (Yeboah et al., 2014).

Social media has various uses in the classroom as well as to help market and promote schools and universities (West, 2019). West (2019) identified the following ways to use social media for education. First, social media provides a smoother, more direct communication tool between students, teachers and parents, who can check in and ask or respond to questions. Social media allows for more e-learning opportunities as well. As remote jobs and online classes are becoming more popular, training students to work from a distance is an important lesson, and social media can help with that. Research has shown that some social media platforms such as WhatsApp as benefits such as the ability to create, share, adapt and reuse content, engage in digital dialogue and collaborate, have peer-to-peer contact, social interaction with other users, its discoverability, and continuous accessibility have attracted greater number of digital

communication users using the platform (Lenhart et al., 2007; Yeboah & Ewur, 2014). Gon and Rawekar (2017) assessed the effectiveness of social media like WhatsApp in delivering knowledge to 4th semester MBBS students and to compare the improvement of knowledge gain through e-learning and didactic lecture. The post test results revealed that the learners with WhatsApp as a learning tool scored in the range of 5–20 marks while those with didactic lecture scored in the range of 3–17.5 marks and had an average of 11.6 and 11.9 score, respectively. Technical, educational, and instructional advantages of teaching learning activity via WhatsApp out pars disadvantages.

Abdulahi et al. (2014) and Oyelere et al. (2016) pointed out that social networking tools offer students the opportunity to communicate, get in touch, access information, research, and chat. The extensive use of social media has encouraged teachers to adopt social media as an e-learning platform (Qi, 2019). According to West (2019) in a visual heavy class, students can use Instagram to present a series of photos or graphics in a visually appealing manner. Instagram allows students to practice digital storytelling in ways that other social media platforms may fall short. West (2019) added that students can create class-specific Instagram accounts and may delete them after the course is over if they so choose. The study by Agbo et al. (2020) found that students in Nigerian higher institutions use WhatsApp to interact and collaborate effectively with their peers and teachers. The authors added that with WhatsApp, students who shy away from seeking face-to-face academic help from teachers and peers have the opportunity to interact remotely.

Lane (2016) provided evidence to suggest that a collaborative learning environment, such as WhatsApp, is satisfying to students and motivates them to continue their learning experience. Kolog et al. (2018) found the use of electronic devices, in the academic environment, as tools for promoting social and collective learning. However, the researchers cautioned that electronic devices used in academic environments must

be used under strict conditions that gear toward learning. There are various uses of social media but more people, particularly students, find it more productive when they use it to learn in groups (Fauzi, 2019). The study by Abbas et al. (2019) reported that results from a Massive Open Online Courses (MOOCs) delivered through social media showed improved performance of the students. Furthermore, it indicated that the involvement of social media and technology in learning programs did help in reducing the number of students who dropped out of the programs. According to West (2019) teachers can create a single Twitter handle per class and reuse it every year, or they can create a new handle each school year. The 280-character limit makes students think critically on communicating concisely and effectively, a beneficial skill to develop. This shows that Twitter can be great as a discussion board or message board for a class. West (2019) pointed out that teachers can use Twitter to post reminders for assignment due dates or share inspirational quotes and helpful links to practice quizzes or resources. He added that teachers can also create discussions and Twitter chats surrounding a specific hashtag that they create.

Getting feedback from students is glorified by Menkhoff et al. (2014) as an important aspect through which an instructor can monitor if students have comprehended the course content taught in class thereby allowing the instructor to identify knowledge gaps and address them on the spot. More importantly, unlike the traditional face-to-face medium of teaching which have some space and time restrictions, in electronic media, the world becomes the classroom, available 24/7, and not confined to Mondays to Fridays (Dzvapatsva et al., 2014). The notion of 24/7 in this context simply suggests that embracing social media enables students and lecturers to interact outside the formal learning time with more flexibility. Whether computer-based or mobile-based, Menkhoff et al. (2014) and Wheeler (2010) identified three usefulness of social media in a university classroom. First, social media supplements blended learning which helps students create positive contextual learning in relation to pedagogical objectives.

Twitter, for example, can be used to engage with students as learning happens during face-to-face classes and; it can also be used to communicate with students when they are at their homes (the period of distance learning). Second, social media affords students to engage in collaborative learning and finally, social media motivates students to post comments or questions to blogs or Twitter about their subject matter. Some previous studies have provided evidence on the benefits and use of Twitter in classroom environments (Chawinga, 2017; Conole & Alevizou, 2010; Jones, 2015; Menkhoff et al. 2014). Jones (2015) used Twitter to teach a literature class for 30 undergraduate students at the University of California that put emphasis on students' participation. It is reported in the study that "my students embraced Twitter as a collaboration tool, and increasingly came to class with improved attitudes toward, and readiness for class discussions" (Jones, 2015, p. 91).

The study by Conole and Alevizou (2010) commissioned by the Higher Education Academy in the United Kingdom highlighted evidence about the benefits of social media in higher education. By drawing data from published reports, conference papers, journal articles and anecdotal evidence, Conole and Alevizou (2010) stated that Twitter and blogs and other Web 2.0 technologies are catalysts of scholarly practice and the sharing of designs and good practice. This is made possible by allowing students to participate in a distributed network of educators and researchers and to co-create knowledge and develop a skill of critique of content (Conole & Alevizou, 2010). In relation to the impact of teacher–student communication through social networks on the teaching–learning process, the improvement of teacher–student communication and teacher–student relationship stands out (Albayrak & Yildirim, 2015; Ean & Lee, 2016). According to West (2019) instructors can create Pinterest boards for each of their classes and save pins that are relevant to lessons. Pinterest is a great social media platform for teachers to use to prepare and organize resources, lesson plans, and worksheets for their classes in one place. Create boards according to class or subject, and create sub-topic boards for

weekly units or all worksheets. Pinterest can also be useful for students to curate a digital bibliography for research projects, papers, or group assignments. Students can pin websites, books, or videos to a board on a single topic and refer back to it when it's time to write.

2.3 CHALLENGES OF USING SOCIAL MEDIA IN INSTRUCTION

Despite the benefits related to the use of social media in the teaching/learning process, some studies suggest that there are various challenges militating against successful integration of most social media in teaching and learning in university environments. For example, Dzvapatsva et al. (2014) at the University of the Western Cape, South Africa, reported that some of the factors that militate against the use of social media include lack of technical skills that students experience when learning or using portals of social media, inadequate technological infrastructure and bandwidth which results in students abusing it.

The study by Conole and Alevizou (2010) revealed a number of issues including privacy where it is reported that there is a lack of understanding of the implications of adopting more open approaches in technological environments, lack of rewards or incentives for instructors using these technologies in class, lack of skills to use these technologies and a belief that these technologies may not necessarily work in a classroom. In the study by Cetinkaya (2017), few students expressed adverse opinions about the timing of some posts and the redundant posts within the group. In some studies, Teachers consider it to be inappropriate for other teachers to harass or mock students, use unauthorized information, propagate false information, create fictitious profiles, share disruptive information, or insult national values on social networks (Asterhan & Rosenberg, 2015; Deveci & Kolburan, 2015).

Some other studies have shown that time students spend during lectures on social media negatively affect student performance. For example, Paul et al. (2012) examined the relationship between the academic performance of 340 business students at a large state university and their use of social media. They found a negative relationship between time spent by students on social media and their academic performance (Paul et al., 2012). Obi et al. (2017) investigated the influence of mobile phone services on students' study habits in tertiary institutions in the South-South and South-East zone of Nigeria. The study revealed that chatting on Facebook, online sport betting, downloading and playing music, WhatsApp chatting, e-mail services are among the mobile phone Internet services used by the students. The study found that using these mobile phone services during private study and lecture time negatively affects students' time management and distracts students' concentration.

Barczyk and Duncan (2011) observe that critics of social media in academia often point out that social networking sites such as Twitter and Facebook are distracters to learning. Consequently, some instructors in higher institutions have been reluctant to adopt social media in their teaching and learning activities (Galagan, 2010). Abbas et al. (2019) warned that despite the overwhelming importance of social media in the everyday life of learners and instructors, users must ensure a healthy balance as it has been reported that social media has both positive and negative impacts on students' learning processes (Obi et al., 2017; van Zoonen et al., 2017). Abbas et al. (2019) investigated the relationship between positive and negative characteristics of social media, and the learning attitude of university students for sustainable education in Pakistan. The study revealed that the use of social media in Pakistan has a negative influence on students' behavior as compared to positive behavior. However, the authors pointed out that the result cannot be generalized beyond the specific context of the research and recommended further research.

According to Fauzi (2019), students' preoccupation with operating smartphones diverts their attention in lectures. He added that based on the observations of researchers during lectures, when there was a presentation in front of the class, students who were sitting behind were comfortable operating a smartphone without paying attention to their friends in front. When entering the discussion session, only the students who sit ahead are actively involved in the discussion. The incident did not only happen once according to the author, but several meetings and repeated. This causes a limited interaction between lecturers and students in the classroom. This has an impact on the lack of activity and their involvement in the discussion, whereas with the discussion activities, they will get a lot of information. The solution to that problem is to develop WhatsApp Group Discussion (Fauzi, 2019).

MICROSOFT TEAMS FOR EDUCATION

Microsoft Teams or MS Teams is primarily used for staff communication and both internal and external events, but it is also available to UCL students and may offer an improved student experience in some situations. Unfortunately it is not currently integrated with Moodle or Portico so managing class groups involves manual effort.

Microsoft Teams for education offers many innovative tools specifically for education which are not available in the business versions of teams.

These educational features include:

Assignments - Assign, track and grade assignments,

Immersive Reader - Easily read conversations and chat in a personalized format

Whiteboard - Collaborate in a freeform, infinite digital canvas

Raise Your Hand - Be heard without disrupting other students

Together Mode - See the whole class in a new way

Accessibility - ASL visibility features and live closed captions

LMS Integrations - Custom educational experiences and LMS integrations

ZOOM FOR EDUCATION

Zoom offers two separate plans for education. Both plans start at $90 per year per user but are differentiated by certain features and software design.

Zoom for Higher Education is used by 96% of top US universities (US News and World Report), while Zoom for K-12 is steadily carving out a market share among schools and districts. Zoom for education boasts additional features to improve classroom and learning performance. Zoom gives you better performance for some students. For online groups larger than 250 participants, Zoom is also a good choice. For smaller groups and 1:1 meetings, MS Teams may offer some advantages. Ideally, it is better to use one platform as much as possible so that you and your students become familiar with it. For basic hybrid teaching that does not require synchronous activities within the same platform, Lecture cast Live is the simplest option for teaching staff, but the least interactive for remote students.

THESE EDUCATIONAL FEATURES INCLUDE:

Session Recording - Record lessons for playback

Student Engagement Tracking - Track student engagement

LMS Integration - Integrate with existing Learning Management Systems

Accessibility - Captioning and keyboard shortcuts

Enhanced Collaboration - One-click content sharing, real-time co-annotation, and digital white boarding

Zoom has been recently acquired by UCL on a 12-month license as it provides greater reliability for staff and students in China. It is also integrated with Moodle.

GOOGLE MEET FOR EDUCATION

G Suite does not have any features specifically for education that differ from their enterprise solution, however, the G Suite platform is built for collaboration and offers a great medium for online learning and teaching.

Google Meet is a fantastic conferencing platform and Google Chat offers a chatting/collaboration/file sharing tool for added collaborating ability.

Using Google Classroom, Google Assignment and Google Forms, educators can create class curriculums, assign projects and grade with ease.

GOOGLE SUITE (MEET) INCLUDE:

Collaboration - Google's entire product suite was built with collaboration in mind

Products - Google Suite appears to have a product for every task

Ease-of-use - If you use Gmail, you are likely already familiar with at least a handful Google products

Open Source - Google offers limitless open source templates for nearly any task

2.4 THEORETICAL FRAMEWORK

Theories and Frameworks for Online Education:

Seeking an Integrated Model

Terry Anderson (2011) examines whether a common theory for online education can be developed. While recognizing that as a difficult, and perhaps fruitless, task, he nonetheless examines possibilities and proposes his own theory which he admits is not complete. The purpose of this article is to examine theoretical frameworks relevant to the pedagogical aspects of online education. It starts with a consideration of learning theories and funnels down to their specific application to online education. The article concludes with a proposal for an integrated model for online education based on pedagogical purpose.

Learning Theory

Learning theory is meant to explain and help us understand how people learn; however, the literature is complex and extensive enough to fill entire sections of a library. It involves multiple disciplines, including psychology, sociology, neuroscience, and of course, education. Three of the more popular learning theories—behaviorism, cognitivist, and social constructivism—will be highlighted to form the foundation for further discussion. Mention will also be made of several other learning theories that are relevant to online education. Before reviewing these theories, it will be worthwhile to have a brief discussion of the term theory itself. Theory is defined as a set of statements, principles, or ideas that relate to a particular subject. A theory usually describes, explains, and/or predicts phenomena. The definition of theory also varies depending upon disciplines, especially when related to the term model. As noted by Graham, Henrie, and Gibbons (2013), the two terms are used interchangeably and generally refer to the same concept. However, a model is more frequently a visual representation of

reality or a concept. In this discussion, the terms theory and model will be used interchangeably. The purpose of a theory or model is to propose the answers to basic questions associated with a phenomenon. Graham, Henrie and Gibbons (2013) reviewed this issue as related to instructional technology and recommended a three-part taxonomy first proposed by Gibbons and Bunderson (2005) that includes theories that: § Explore: "What exists?" and attempts to define [describe] and categorize; § Explain: "Why does this happen?" and looks for causality and correlation, and work with variables and relationships. § Design: "How do I achieve this outcome?" and describes interventions for reaching targeted outcomes and operational principles (Graham, Henrie and Gibbons, 2013, p. 13). This taxonomy will serve as an overall guiding principle for the discussion of learning theories and models in this article.

Behaviorism

As its name implies, behaviorism focuses on how people behave. It evolved from a positivist worldview related to cause and effect. In simple terms, action produces reaction. In education, behaviorism examines how students behave while learning. More specifically, behaviorism focuses on observing how students respond to certain stimuli that, when repeated, can be evaluated, quantified, and eventually controlled for each individual. The emphasis in behaviorism is on that which is observable and not on the mind or cognitive processes. In sum, if you cannot observe it, it cannot be studied. The development of behaviorism is frequently associated with Ivan Pavlov, famous for his experiments with dogs, food, and audible stimuli, such as a bell. In his experiments, dogs learned to associate food or feeding time with the sound of the bell and began to salivate. Pavlov conducted his experiments in the early 1900s and they were replicated by many other researchers throughout the 20th century. John B. Watson, among the first Americans to follow Pavlov's work, saw it as a branch of natural science. Watson

became a major proponent of Pavlov and is generally credited with coining the term behaviorism. He argued that mind and consciousness are unimportant in the learning process and that everything can be studied in terms of stimulus and response. Other major figures associated with behaviorism are B.F. Skinner and Edward Thorndike. Skinner is particularly well known, primarily because he introduced what he referred to as operant conditioning which emphasized the use of both positive and negative reinforcement to help individuals learn new behaviors. This was quite different from Pavlov, who relied on simple reflexive responses to specific stimuli although both Pavlov and Skinner promoted repetitive behavior that leads to habit formation. Skinner had a significant influence on early computer assisted instructional (CAI) models as developed by Pat Suppes and others. A common aspect of early CAI programs was the reliance on encouragement and repetition to promote positive learning activities.

Behaviorism led to the development of taxonomies of learning because it emphasized the study and evaluation of multiple steps in the learning process. Behaviorists repeatedly studied learning activities to deconstruct and define the elements of learning. Benjamin Bloom (1956) was among the early psychologists to establish a taxonomy of learning that related to the development of intellectual skills and to stress the importance of problem solving as a higher order skill. Bloom's (1956) Taxonomy of educational objectives handbook: Cognitive domains remains a foundational text and essential reading within the educational community. Bloom's taxonomy is based on six key elements (see Figure 1) as follows:

Creating: Putting elements together to form a coherent or functional whole, and reorganizing elements into a new pattern or structure through generating, planning, or producing.

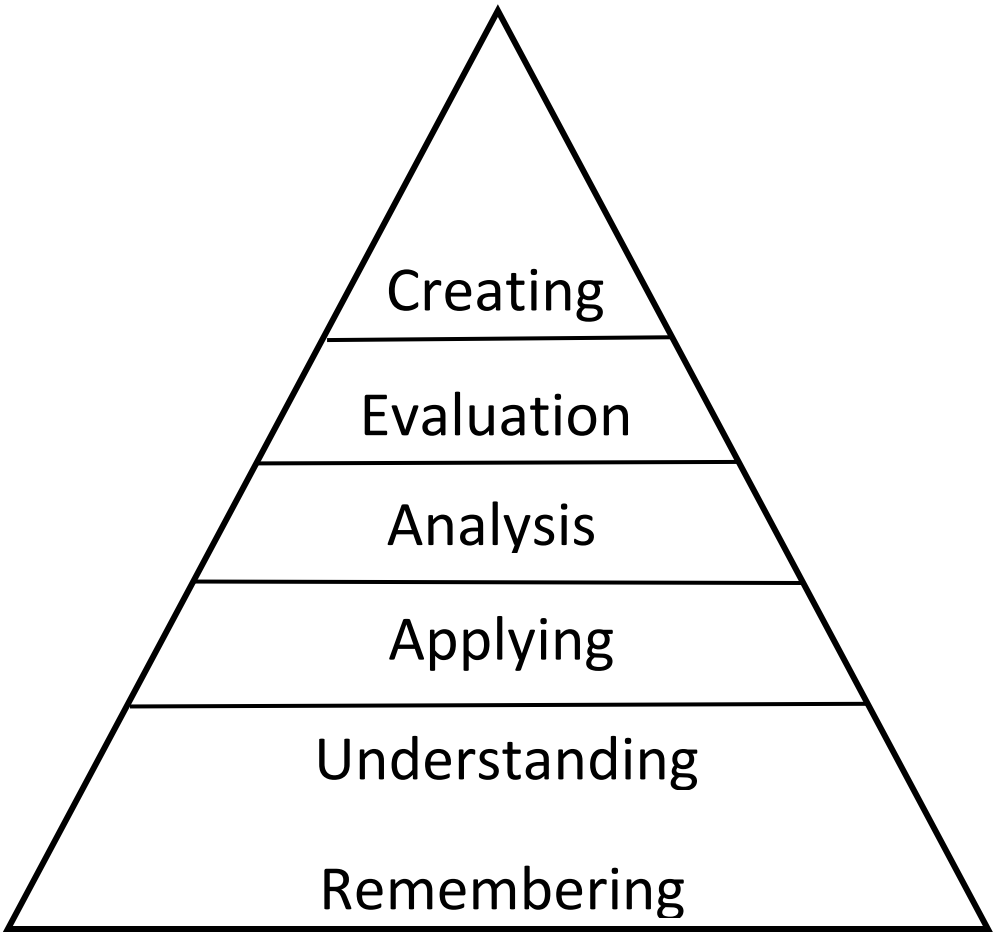
Evaluating: Making judgments based on criteria and standards through checking and critiquing.

Analyzing: Breaking material into constituent parts, and determining how the parts relate to one another and to an overall structure or purpose through differentiating, organizing, and attributing.

Applying: Carrying out or using a procedure through executing or implementing.

Understanding: Constructing meaning from oral, written, and graphic messages through interpreting, exemplifying, classifying, summarizing, inferring, comparing, and explaining.

Remembering: Retrieving, recognizing, and recalling relevant knowledge from long-term memory



Bloom, in developing his taxonomy, essentially helped to move learning theory toward issues of cognition and developmental psychology. Twenty years later, Robert Gagne, an

educational psychologist, developed another taxonomy (events of instruction) that built on Bloom's and became the basis for cognitivist instructional design (Harasim, 2012). Gagne emphasized nine events in instruction that drive the definitions of objectives and strategies for the design of instructional material.

1. Gain attention: Use media relevant to the topic.

2. Describe the goal: Provide clear objectives to the overall course goals.

3. Stimulate prior knowledge: Review previously presented material and concepts and connect them to the material to be addressed in the current module.

4. Present the material to be learned: Readings, presentations, demonstrations, multimedia, graphics, audio files, animations, etc.

5. Provide guidance for learning: Discussions to enable learners to actively reflect on new information in order to check their knowledge and understanding of content.

6. Elicit performance: Activity-based learning such as group research projects, discussion, homework, etc.

Cognitivist

Cognitivist has been considered a reaction to the "rigid" emphasis by behaviorists on predictive stimulus and response (Harasim, 2012, p. 58). Cognitive theorists promoted the concept that the mind has an important role in learning and sought to focus on what happens in between the occurrence of environmental stimulus and student response. They saw the cognitive processes of the mind, such as motivation and imagination, as critical elements of learning that bridge environmental stimuli and student responses. For example, Noam Chomsky (1959) wrote a critical review of Skinner's behaviorist work

in which he raised the importance of creative mental processes that are not observable in the physical world. Although written mainly from the perspective of a linguist, Chomsky's view gained popularity in other fields, including psychology. Interdisciplinary in nature, cognitive science draws from psychology, biology, neuroscience, computer science, and philosophy to explain the workings of the brain as well as levels of cognitive development that form the foundation of learning and knowledge acquisition. As a result, cognitivist has evolved into one of the dominant learning theories. The future of cognitivist is particularly interesting as more advanced online software evolves into adaptive and personalized learning applications that seek to integrate artificial intelligence and learning analytics into instruction.

Chapter Three

Research Methodology

3.1 Research Design

The study adopted survey design. This is because the study sought for opinion of sample of pre-service teachers in three tertiary institutions of learning in Lagos State

3.2 Population of the study

The target population for this study comprises of undergraduates in Yaba and Mainland Local Government area of Lagos state. The population will consist of undergraduates from University of Lagos, Undergraduates from University of Ibadan in affiliation with St Augustine College of Education Akoka, and Undergraduates from University of Benin in affiliation with FCE(T) Akoka.

3.3 Sample and Sampling Techniques

The sample comprised of 600 undergraduates randomly selected. 200 students from the University of Benin in affiliation with FCE(T) Akoka, 300 students from University of Lagos, 100 students from University of Ibadan in affiliation with St Augustine College of Education Akoka. Stratified Random Sampling Techniques was used for the sample selection. The total sample size is 600 students.

3.4 Instrument for Data Collection

The research instrument that was used in the study is questionnaire. The questionnaire was personally designed by the researcher based on the purpose of the study and research questions. The questionnaire comprised of two major sections labeled A and B Section. A sought for bio-data information of respondent students such as sex, age, institution and discipline, All the items in section A are close ended with a stem and possible options.

On the other hand, selection B of the questionnaire consisted of 15 items divided into subsections. Each subsection was designed to collect information about specific questions. The questions are all structured. Each has an option statement and four alternative responses to which the respondents are only to tick the one that is most application or relevant to them. The options include:

SA Strongly Agree

A Agree

D Disagree

SD Strongly Disagree

3.5 Validation of Instrument

The questionnaire was validated before being used for data collection. This was done when three draft copies of it were given to senior lecturers of University of Benin and Federal College of Education (Technical) Akoka. In fact, they advised the researcher on item that are irrelevant in terms of their content, construct and

degree of relatedness to the purpose of study. They also recommended some areas of adjustment and addition in some cases. These corrections were effected by the researcher. These exercises helped to ensure that the questionnaire had high content and construct validates.

3.6 Reliability of the Instrument

Before being used for data collection, the questionnaire was trial tested as a means of ensuring the reliability. Ten copies of the validated ones were administered on ten respondents' undergraduates In St Augustine College of Education a private college of Education which was not part of the main study. The data resulting from the pilot study was analyzed as a means of determining the reliability coefficient of 0.79. This high coefficient is an indication that the instrument had consistency and high reliability.

3.7 Method of Data Collection

The final forms of the questionnaire were partly administered by the researcher and partly administered by two (2) research assistants trained for this purpose. These students were undergraduates of the institutions sampled. The data collection exercise lasted for two weeks.

3.8 Method of Data Analysis

The data collected for the study were analyzed with frequency count, percentage (%) Mean; and Standard Deviation.

Any item with the mean value of 3.50 or above was considered as not required.

FEDERAL COLLEGE OF EDUCATION TECHNICAL IN AFFILIATION WITH UNIVERSITY OF BENIN

Questionnaire on the evaluation of undergraduate use of selected social networks for instruction in Lagos state

This questionnaire is designed to elicit information for the evaluation of undergraduate use of selected social networks for instruction in Lagos state.

Students Questionnaire

Please kindly tick (√) your rating and write general comments.

Section A:

Sex: Female ( ) Male ( )

Institution: UNIBEN( ) UNILAG ( ) UNI ILORIN ( )

Level: 100 ( ) 200 ( ) 300 ( ) 400 ( )

SECTION B:

Please kindly write **SA –** Strongly Agree, **A –** Agree, **SD –** Strongly Disagree and **D –** Disagree.

Undergraduate awareness about the availability of social networks for instruction

S/N	**Statements**	**Strongly Agree**	**Agree**	**Strongly Disagree**	**Disagree**
1	I know that the Zoom app can be used				

	for lecture				
2	I know Mocrosoft teams app can be used for lecture				
3	I know the Google Meet app can be used for lecture				

Various social networks undergraduate use for instruction

S/N	**Statements**	**Strongly Agree**	**Agree**	**Strongly Disagree**	**Disagree**
1	We use the Zoom App for our Lectures				
2	We use the Microsoft Teams App for our lecturers				
3	We use the Google Meet App for our lecturers				

The frequency of use of available social network for undergraduate instruction

S/N	Statements	Strongly Agree	Agree	Strongly Disagree	Disagree
1	We use the Zoom App for our Lectures every semester				
2	We use the Microsoft Teams App for our lecturers every semester				
3	We use the Google Meet App for our lecturers every semester				

Undergraduates understanding of the functions in using social networks for instruction

S/N	Statements	Strongly Agree	Agree	Strongly Disagree	Disagree
1	I know how to use mute and unmute on the zoom app				
2	I know how to share slides on the Google Meet App				
3	I know how to raise my hands on the Microsoft Teams App				

Undergraduate challenges in using of social networks for instruction

S/N	Statements	Strongly Agree	Agree	Strongly Disagree	Disagree
1	I always have data challenge when using the Zoom App				
2	I always have network challenge when using Google Meet App				
3	I am always logged out when using the Microsoft Teams App				

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